

TidyTuesday Week 40

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Wednesday 30th September, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly ritual where we take two economic series that have never met, force them into a linear regression, and then act astonished by whatever falls out. This week the random number generator served up wood product manufacturing in the Great Lakes, European cell phone subscriptions, discontinued wage-share statistics, Black Americans not in the labor force, intellectual property import charges, and the durable-goods slice of consumer spending. A truly harmonious guest list, all of whom got along about as well as you'd expect strangers to.

Monday's matchup pitted Great Lakes wood product GDP against mobile cellular subscriptions in Europe and Central Asia, which is exactly the kind of pairing this project exists to humiliate. The raw regression shrugged – not even significant, a rare case of two series honestly admitting they have nothing to say to each other. But then, plot twist nobody asked for, the detrended version actually crossed the significance line. So after we strip out the trends, the leftover wiggles in Midwestern lumber output and the leftover wiggles in Eurasian phone plans quietly agree on something. Is it real? Almost certainly not. But it's the one result this week that got stronger after detrending, which technically makes it the most "interesting," in the way that finding a sock behind your dryer is technically an achievement.

Tuesday delivered the week's big drama queen: a discontinued measure of the wage-and-salary share of gross domestic income versus the number of Black Americans not in the labor force. The raw regression posted an R-squared of 0.89 and a p-value with more zeros than a lottery jackpot, the sort of number that makes you briefly believe you've discovered a Law of Economics. Then detrending walked in, calmly removed the shared drift these two long series had been coasting on, and the relationship deflated from "world-changing" to a much humbler still-significant-but-please-calm-down. It's the textbook trend mirage: most of that gorgeous fit was just two series wandering in the same general direction across decades, holding hands purely because time was passing.

Wednesday's pairing of intellectual property import charges against the durable-goods share of consumer spending did the same striptease – a respectable raw R-squared of 0.32 and a significant p-value that promptly evaporated the instant we detrended, landing at a p-value of 0.44 that says, in no uncertain terms, "nothing to see here." As always, the standing reminder: these are randomly paired series, run through the bluntest possible statistics, with zero causal claims and even less dignity. Spurious correlation isn't the bug here, it's the entire personality of the project. See you next week for more of it.

2 Date: 2026-09-28

Series ID: GLAKWPMANRGSP

This series is titled Real Gross Domestic Product: Wood Product Manufacturing (321) in the Great Lakes BEA Region and has a frequency of Annual. The units are Millions of Chained 2017 Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1997-01-01 and the observation end date is 2024-01-01. The popularity of this series is 1.

Series ID: ITCELSETSP2ECS

This series is titled Mobile Cellular Subscriptions: All Income Levels in Europe and Central Asia and has a frequency of Annual. The units are Number per 100 People and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1960-01-01 and the observation end date is 2025-01-01. The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_ITCELSETSP2ECS	R-squared:	0.083
Model:	OLS	Adj. R-squared:	0.048
Method:	Least Squares	F-statistic:	2.357
Date:	Mon, 28 Sep 2026	Prob (F-statistic):	0.137
Time:	18:50:21	Log-Likelihood:	-142.11
No. Observations:	28	AIC:	288.2
Df Residuals:	26	BIC:	290.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	285.3207	123.888	2.303	0.030	30.665	539.976
value_fred_GLAKWPMANRGSP	-0.0407	0.026	-1.535	0.137	-0.095	0.014

Omnibus:	3.492	Durbin-Watson:	0.174
Prob(Omnibus):	0.174	Jarque-Bera (JB):	2.983
Skew:	-0.713	Prob(JB):	0.225
Kurtosis:	2.275	Cond. No.	7.63e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 7.63e+04. This might indicate that there are strong multicollinearity or other numerical problems.

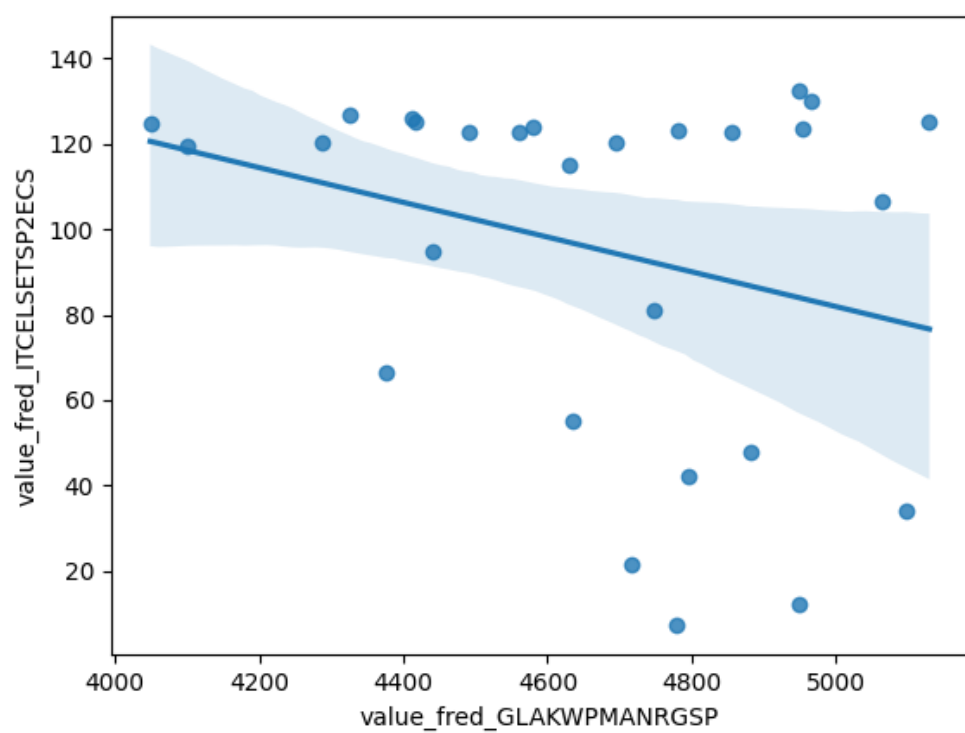


Figure 1: Raw Regression Plot for 2026-09-28

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_ITCELSETSP2ECS_detrended	R-squared:	0.184
Model:	OLS	Adj. R-squared:	0.152
Method:	Least Squares	F-statistic:	5.853
Date:	Mon, 28 Sep 2026	Prob (F-statistic):	0.0228
Time:	18:50:21	Log-Likelihood:	-119.03
No. Observations:	28	AIC:	242.1
Df Residuals:	26	BIC:	244.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.116e-12	3.330	3.35e-13	1.000	-6.845	6.845
value_fred_GLAKWPMANRGSP_detrended	-0.0283	0.012	-2.419	0.023	-0.052	-0.004

Omnibus:	0.861	Durbin-Watson:	0.444
Prob(Omnibus):	0.650	Jarque-Bera (JB):	0.837
Skew:	0.216	Prob(JB):	0.658
Kurtosis:	2.272	Cond. No.	285.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

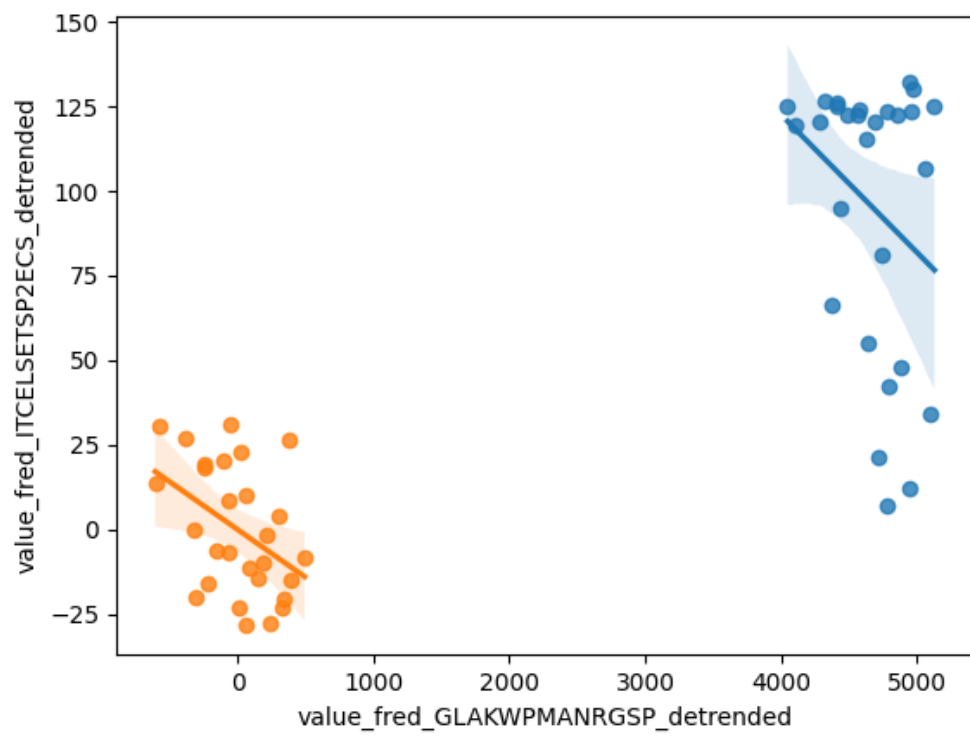


Figure 2: Detrended Regression Plot for 2026-09-28

3 Date: 2026-09-29

Series ID: W269RE1A156NBEA

This series is titled Shares of gross domestic income: Compensation of employees, paid: Wage and salary accruals: Disbursements (DISCONTINUED) and has a frequency of Annual. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1929-01-01 and the observation end date is 2011-01-01. The popularity of this series is 1.

Series ID: LNU05000006

This series is titled Not in Labor Force, Black or African American and has a frequency of Monthly. The units are Thousands of Persons and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1972-01-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_LNU05000006	R-squared:	0.886			
Model:	OLS	Adj. R-squared:	0.883			
Method:	Least Squares	F-statistic:	295.5			
Date:	Tue, 29 Sep 2026	Prob (F-statistic):	1.63e-19			
Time:	17:07:32	Log-Likelihood:	-300.21			
No. Observations:	40	AIC:	604.4			
Df Residuals:	38	BIC:	607.8			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t 	[0.025	0.975]
const	3.833e+04	1754.440	21.850	0.000	3.48e+04	4.19e+04
value_fred_W269RE1A156NBEA	-632.0219	36.768	-17.190	0.000	-706.454	-557.590
Omnibus:	2.321	Durbin-Watson:	0.414			
Prob(Omnibus):	0.313	Jarque-Bera (JB):	1.433			
Skew:	0.178	Prob(JB):	0.488			
Kurtosis:	2.143	Cond. No.	1.17e+03			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.17e+03. This might indicate that there are strong multicollinearity or other numerical problems.

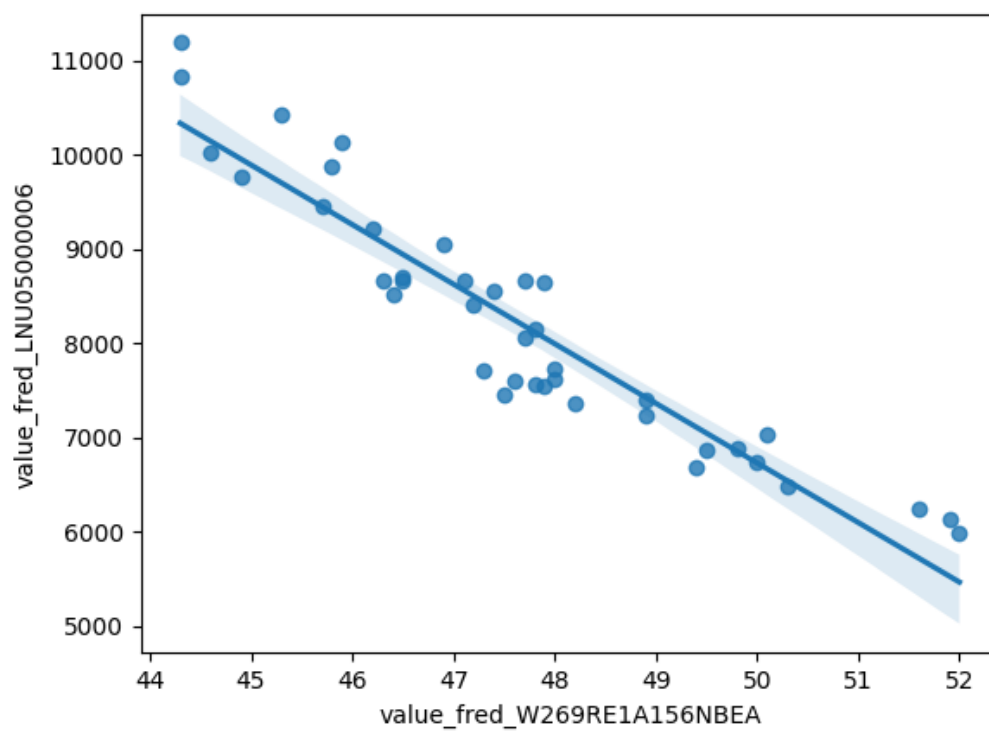


Figure 3: Raw Regression Plot for 2026-09-29

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_LNU05000006_detrended	R-squared:	0.229
Model:	OLS	Adj. R-squared:	0.209
Method:	Least Squares	F-statistic:	11.29
Date:	Tue, 29 Sep 2026	Prob (F-statistic):	0.00178
Time:	17:07:32	Log-Likelihood:	-275.80
No. Observations:	40	AIC:	555.6
Df Residuals:	38	BIC:	559.0
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.404e-11	38.755	1.14e-12	1.000	-78.455	78.455
value_fred_W269RE1A156NBEA_detrended	-174.7065	51.991	-3.360	0.002	-279.957	-69.456

Omnibus:	8.056	Durbin-Watson:	0.312
Prob(Omnibus):	0.018	Jarque-Bera (JB):	6.850
Skew:	0.862	Prob(JB):	0.0325
Kurtosis:	4.067	Cond. No.	1.34

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

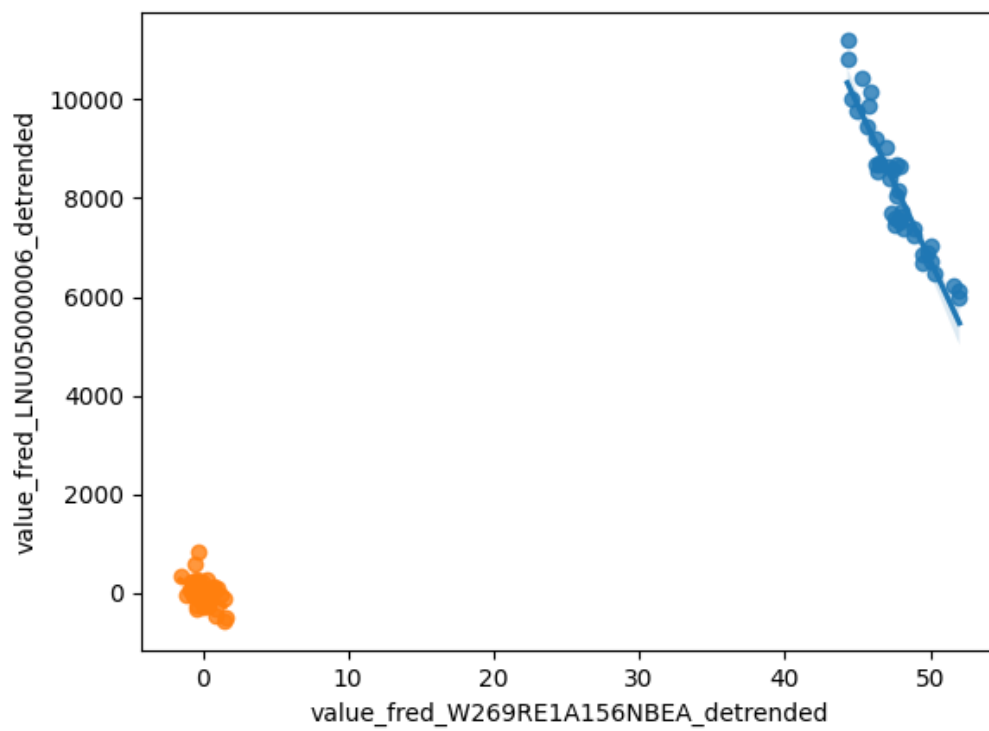


Figure 4: Detrended Regression Plot for 2026-09-29

4 Date: 2026-09-30

Series ID: IEAMSIPN

This series is titled Imports of Services: Charges for the use of intellectual property n.i.e. and has a frequency of Quarterly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1999-01-01 and the observation end date is 2026-04-01. The popularity of this series is 1.

Series ID: DDURRE1A156NBEA

This series is titled Shares of gross domestic product: Personal consumption expenditures: Durable goods and has a frequency of Annual. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1929-01-01 and the observation end date is 2025-01-01. The popularity of this series is 4.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_DDURRE1A156NBEA	R-squared:	0.321
Model:	OLS	Adj. R-squared:	0.294
Method:	Least Squares	F-statistic:	11.82
Date:	Wed, 30 Sep 2026	Prob (F-statistic):	0.00206
Time:	17:05:45	Log-Likelihood:	-25.239
No. Observations:	27	AIC:	54.48
Df Residuals:	25	BIC:	57.07
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	8.7930	0.314	28.002	0.000	8.146	9.440
value_fred_IEAMSIPN	-0.0001	3.39e-05	-3.438	0.002	-0.000	-4.67e-05

Omnibus:	5.919	Durbin-Watson:	0.272
Prob(Omnibus):	0.052	Jarque-Bera (JB):	1.819
Skew:	0.005	Prob(JB):	0.403
Kurtosis:	1.729	Cond. No.	2.36e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.36e+04. This might indicate that there are strong multicollinearity or other numerical problems.

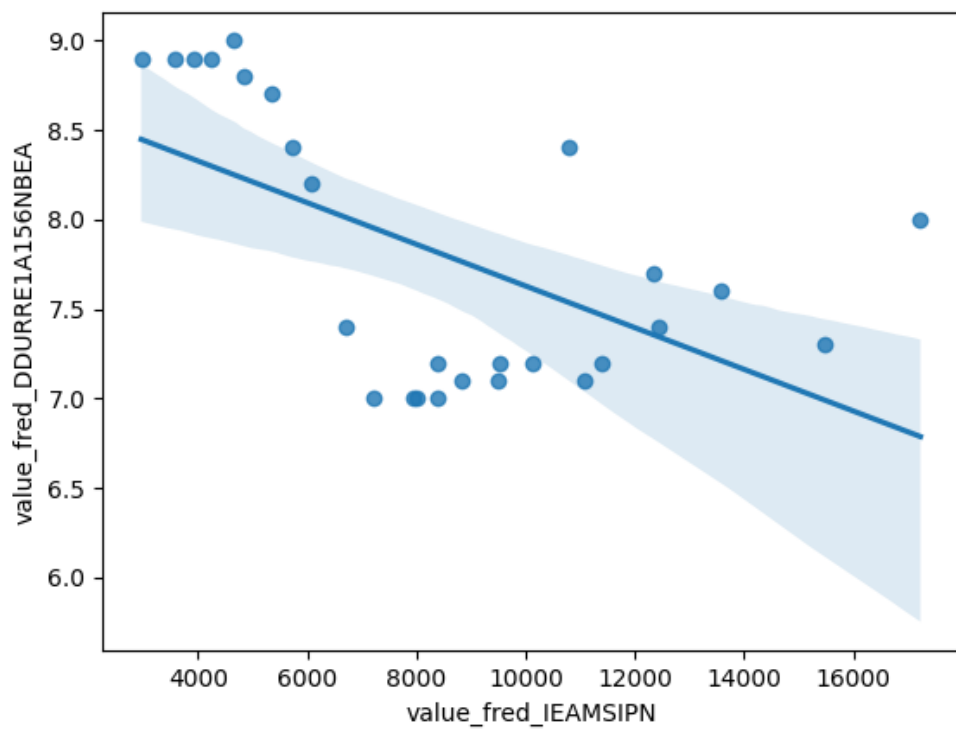


Figure 5: Raw Regression Plot for 2026-09-30

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_DDURRE1A156NBEA_detrended	R-squared:	0.024
Model:	OLS	Adj. R-squared:	-0.015
Method:	Least Squares	F-statistic:	0.6106
Date:	Wed, 30 Sep 2026	Prob (F-statistic):	0.442
Time:	17:05:45	Log-Likelihood:	-23.256
No. Observations:	27	AIC:	50.51
Df Residuals:	25	BIC:	53.10
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	6.162e-14	0.115	5.38e-13	1.000	-0.236	0.236
value_fred_IEAMSIPN_detrended	8.159e-05	0.000	0.781	0.442	-0.000	0.000

Omnibus:	0.623	Durbin-Watson:	0.358
Prob(Omnibus):	0.732	Jarque-Bera (JB):	0.652
Skew:	-0.067	Prob(JB):	0.722
Kurtosis:	2.250	Cond. No.	1.10e+03

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 1.1e+03. This might indicate that there are strong multicollinearity or other numerical problems.

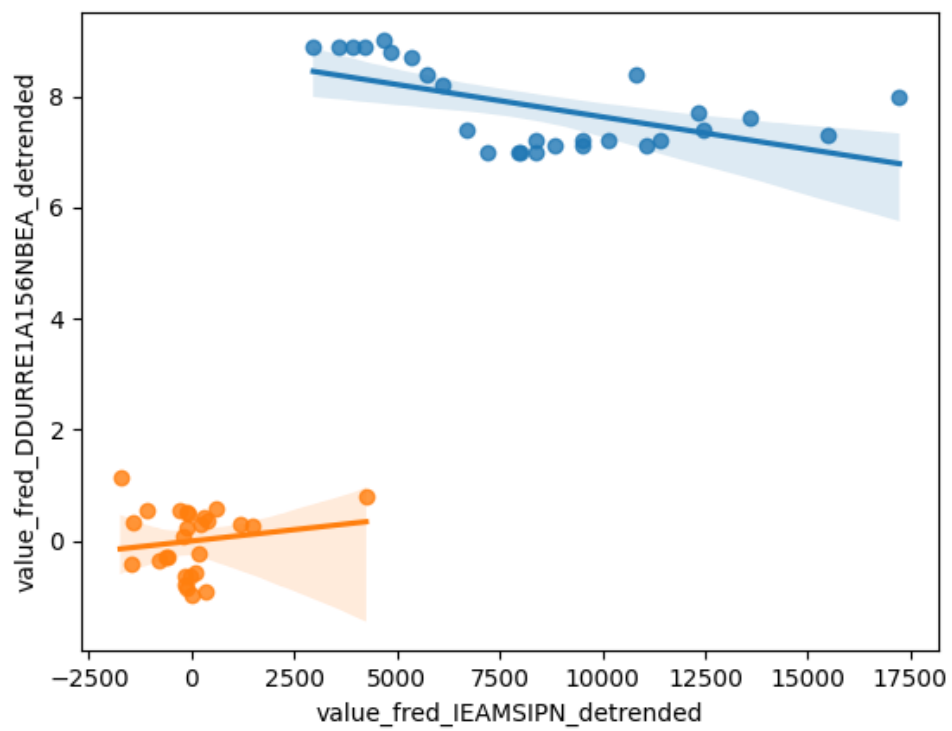


Figure 6: Detrended Regression Plot for 2026-09-30